

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16**COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Heisenberg Raja G B certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Heisenberg Raja

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID Symptoms Observed

P1 Fever,Cough

P2 Fever,Headache

P3 Cough,Headache

P4 Fever,Cough, Headache

P5 Fever,Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID Courses Enrolled

S1 Python, SQL

S2 Python, Data Science

S3 SQL, Machine Learning

S4 Python, SQL, Data Science

S5 Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID Books Borrowed

T1 Data Mining, Python Programming

T2 Data Mining, Database Systems

T3 Python Programming, Database Systems

T4 Data Mining, Python Programming, Database Systems

T5 Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy**Student ID Courses Taken**

S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records**CID Age Group Income Level Policy Purchased**

C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Q1. Dataset: Hospital Patient Records (Min Support = 50%, Min Confidence = 70%, Constraint: all rules must include "Fever")

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Step 1 — Single-item support (Total transactions = 5; 50% support $\Rightarrow$ count ≥ 2.5 , i.e. ≥ 3)

Item	Count	Support %	Frequent ($\geq 50\%$)?
Fever	4	80%	Yes
Cough	3	60%	Yes
Headache	3	60%	Yes
Fatigue	1	20%	No

Step 2 — 2-itemsets containing Fever (constraint applied)

Itemset	Transactions	Count	Support %	Frequent?
Fever,Cough	P1, P4	2	40%	No
Fever,Headache	P2, P4	2	40%	No
Fever,Fatigue	P5	1	20%	No

Result and Justification

No 2-itemset containing "Fever" reaches the 50% minimum support threshold — the highest support among Fever-pairs is 40% (Fever & Cough, Fever & Headache), which falls short of the required 2.5/5 (50%) count. Since a

valid association rule requires its underlying itemset to first be frequent, and none of the Fever-containing itemsets qualify, no association rule can be generated that satisfies both the minimum support (50%) and the constraint of including "Fever".

This demonstrates constraint-based pruning in practice: applying an item constraint ("must include Fever") before confidence evaluation immediately eliminates all candidate rules once the support test fails, so the 70% confidence threshold is never even reached. The exercise illustrates that constraints can legitimately result in an empty rule set — a valid and expected outcome when the data does not support the specified pattern strength.

Q2. Dataset: Online Learning Platform (Min Support = 40%, Constraint: itemsets must contain ≥ 2 courses)

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Step 1 — Item counts (reference only; single items excluded by the ≥ 2 -course constraint)

Item	Count	Support %
Python	4	80%
SQL	3	60%
Data Science	2	40%
Machine Learning	2	40%

Step 2 — Candidate 2-itemsets (min support 40% $\Rightarrow$ count ≥ 2)

Itemset	Transactions	Count	Support %	Frequent?
Python,SQL	S1, S4	2	40%	Yes
Python,Data Science	S2, S4	2	40%	Yes
Python, Machine Learning	S5	1	20%	No
SQL, Data Science	S4	1	20%	No
SQL, Machine Learning	S3	1	20%	No

Step 3 — 3-itemsets

{Python, SQL, Data Science}: only S4 $\rightarrow$ count = 1 (20%) $\rightarrow$ Not frequent.

Frequent Itemsets (satisfying support $\geq 40\%$ and ≥ 2 courses)

- {Python, SQL} — support 40%
- {Python, Data Science} — support 40%

How the Constraint Helps Pruning

The "at least two courses" constraint is an anti-monotone-style item-count constraint that can be pushed directly into the candidate-generation step of Apriori rather than applied only after mining. It removes all single-item itemsets from consideration up front, so the algorithm never wastes computation evaluating or joining 1-itemsets that cannot satisfy the business requirement. During the Apriori join step, only itemsets of size ≥ 2 are generated and support-counted, which shrinks the candidate space, reduces the number of database scans needed to validate candidates, and focuses the mining process on multi-course combinations that are directly actionable for recommending course bundles or pathways.

Q3. Dataset: Library Borrowing Records (Apriori algorithm, Constraint: maximum itemset size = 3)

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Assumption: since the question does not restate a support value, the standard minimum support used across this assessment's other Apriori-style questions is applied — min support = 40% (count ≥ 2 out of 5 transactions).

Step 1 — Frequent 1-itemsets (L1)

Item	Count	Support %	Frequent?
DataMining	4	80%	Yes
Python Programming	3	60%	Yes
Database Systems	3	60%	Yes
Artificial Intelligence	1	20%	No

Step 2 — Candidate & Frequent 2-itemsets (L2), generated by self-joining L1

Itemset	Transactions	Count	Support %	Frequent?
Data Mining, Python Programming	T1, T4	2	40%	Yes
Data Mining, Database Systems	T2, T4	2	40%	Yes
Python Programming, Database Systems	T3, T4	2	40%	Yes

Step 3 — Candidate 3-itemset (L3), generated by joining L2

{Data Mining, Python Programming, Database Systems}: only T4 → count = 1
(20%) → Not frequent.

Effect of the "Maximum Itemset Size = 3" Constraint

Because L3 already fails the minimum support test, the algorithm would terminate naturally at this point even without the constraint. The size constraint's real value shows up as a hard upper bound written into the candidate-generation logic itself: at level 4, Apriori would normally attempt to self-join L3 to build 4-itemset candidates. With the maximum-size constraint in place, the algorithm skips candidate generation beyond size 3 entirely — it does not even attempt the join, prune, and support-count operations for level 4. This directly bounds the itemset lattice that must be traversed, reducing the number of candidate generations, database scans, and support-counting passes, which is especially valuable on larger/denser datasets where higher-order itemsets would otherwise explode combinatorially.

Final Frequent Itemsets

- Size 1: {Data Mining}, {Python Programming}, {Database Systems}
- Size 2: {Data Mining, Python Programming}, {Data Mining, Database Systems}, {Python Programming, Database Systems}
- Size 3: none (below minimum support)

Q4. Dataset: University Course Hierarchy (Multilevel Association Rule Mining, Constraint: rules only at "Computer Science" level)

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Concept Hierarchy

- Level 1 (root/general): Engineering
- Level 2 (mid-level): Computer Science, Information Technology
- Level 3 (specific): Machine Learning (specialization under Computer Science)

Applying the Constraint

- Restricting rule generation to the "Computer Science" level means Engineering (too general — nearly every student has it, so any rule involving it is trivial and uninformative) and Information Technology (a sibling branch, unrelated to CS) are excluded. Mining is instead focused on Computer Science and its descendant, Machine Learning.
- Students taking Computer Science: S1, S3, S4 → 3 of 5 (support 60%).
- Students taking both Computer Science and Machine Learning: S1, S4 → 2 of 5 (support 40%).

Rule	Support	Confidence
Computer Science → Machine Learning	$2/5 = 40\%$	$2/3 = 66.7\%$
Machine Learning → Computer Science	$2/5 = 40\%$	$2/2 = 100\%$

How the Hierarchy Affects Rule Discovery

Multilevel mining lets rules be generated at different levels of abstraction (general to specific) using the same underlying transactions. Without a level constraint, the algorithm would also surface a rule like "Engineering → Computer Science" (support 60%, confidence 60%) — but this is largely uninformative because Engineering is the common parent of nearly all students. By constraining discovery to the Computer Science level, the analysis avoids these redundant, overly general parent-level rules and instead surfaces the specific, actionable pattern that Computer Science students frequently also study Machine Learning — with a particularly strong signal in the Machine Learning → Computer Science direction (100% confidence), useful for curriculum planning or targeted advising.

Q5. Dataset: Insurance Customer Records (Multidimensional Association Rule Mining, Constraint: Age Group = "Young")

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Step 1 — Apply the Age Group = "Young" constraint

Filtering to Young customers leaves C1, C3, and C5 (3 of 5 records). This is a multidimensional analysis because two independent dimensions — Income Level and Policy Type — are examined jointly, conditioned on the Age dimension fixed at "Young".

CID	Income Level	Policy Purchased
C1	High	Health Insurance
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

2 — Patterns among Young customers

Rule	Support (of Young subset, n=3)	Confidence
Age=Young $\wedge$ Income=High $\rightarrow$ Policy=Health Insurance	1/3 = 33.3%	1/1 = 100%
Age=Young $\wedge$ Income=Low $\rightarrow$ Policy=Travel Insurance	1/3 = 33.3%	1/1 = 100%

Age=Young $\wedge$ Income=Medium $\rightarrow 1/3 = 33.3\%$ Policy=Vehicle Insurance		1/1 = 100%
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Interpretation and Marketing Value

Within the Young segment, each income band maps cleanly to a distinct policy type in this sample: high-income young customers align with Health Insurance, low-income young customers with Travel Insurance, and medium-income young customers with Vehicle Insurance. While the dataset is small (each pattern is supported by a single record, so these should be treated as indicative rather than statistically robust), the multidimensional view — combining Age with Income to predict Policy — is exactly the kind of pattern insurers use for micro-segmented, targeted marketing: for example, promoting premium health-cover add-ons to young, high-income customers, budget-friendly travel cover to young, lower-income customers, and vehicle-insurance bundles/discounts to young, medium-income customers, rather than sending one generic campaign to the entire Young demographic.